Part A [15 marks]

Question 1 [15 marks]

(a) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) Suppose $V \sim \chi^2_1$ and $W \sim \chi^2_1$, independently. What is the distribution of $V + W$?

A $G(0, 1)$ approximately

B Z^2 where $Z \sim G(0, 1)$

C Exponential(2) [This is the same as χ^2_2]

D None of the above.

(ii) We should first define the study population during which step of PPDAC?

A Problem

B Plan

C Data

D Analysis

(iii) Suppose $Y \sim$ Poisson(θ). Which of these statements is TRUE?

A θ is a constant.

B θ is a random variable.

C The true value of θ depends on the observed data.

D More than one of the above statements is true.

(iv) A 95% confidence interval for an unknown parameter θ is $[2.3, 4.6]$. Which of the following statements is TRUE?

A $P(2.3 \leq \theta \leq 4.6) = 0.95$

B We are 95% confident $\theta \in [2.3, 4.6]$

C Both A and B are true.

D Neither A nor B is true.

(v) When carrying out a hypothesis test using the likelihood ratio test statistic and a sample of size n, which probability distribution do we use?

A t_n

B t_{n-1}

C χ^2_1

D χ^2_{n-1}

(b) [5 marks] The time (in minutes) spent studying for a STAT 231 exam is assumed to follow a $G(\mu, \sigma)$ distribution. A survey of 20 randomly selected students recorded each student's study time for a STAT 231 exam and observed the following:

$$\sum_{i=1}^{20} y_i = 5000 \quad \text{and} \quad \sum_{i=1}^{20} (y_i - \bar{y})^2 = 17100$$

where y_i = the study time in minutes for the i^{th} student, $i = 1, 2, \ldots, 20$.

You are given the following R commands and output:

```
> qt(0.9, 19)          > qt(0.95, 19)          > qt(0.975, 19)
[1] 1.327728           [1] 1.729133            [1] 2.093024
> qt(0.9, 20)          > qt(0.95, 20)          > qt(0.975, 20)
[1] 1.325341           [1] 1.724718            [1] 2.085963
> qnorm(0.9)           > qnorm(0.95)           > qnorm(0.975)
[1] 1.281552           [1] 1.644854            [1] 1.959964
```

(i) [1 mark] Write down numeric expressions for the sample standard deviation s and the maximum likelihood estimate of the sample mean $\hat{\mu}$.

$$\hat{\mu} = \bar{y} = \frac{5000}{20} = 250$$

$$s = \sqrt{\frac{1}{19} \sum_{i=1}^{20} (y_i - \bar{y})^2} = \sqrt{\frac{17100}{19}} = \sqrt{900} = 30$$

[0.5 marks each]

(ii) [2 marks] Construct a **90%** confidence interval for μ. Your answer may be left as a numeric expression, but you must clearly indicate the numeric values of any terms used in the expression.

The CI is given by $\hat{\mu} \pm a \frac{s}{\sqrt{n}}$ where $a = 1.729$

[1 mark for correct equation $\hat{\mu} \pm a \frac{s}{\sqrt{n}}$. 1 mark for correct quantile.]

(iii) [2 marks] A **90%** confidence interval for σ^2 is given by the following equation:

$$\left(\frac{(n-1)s^2}{b}, \frac{(n-1)s^2}{a} \right)$$

Fill in the blanks in the following statement so that a and b are appropriately defined:

$P(W \leq a) = \underline{0.05}$ and $P(W \leq b) = \underline{0.95}$

where W follows a chi-squared distribution with $\underline{19}$OR$n - 1$ degrees of freedom.

0.5 marks for each space. **Note**: different values for a and b are possible.

(c) [5 marks] Riley and Sam own a cat called Joey. When Riley and Sam watch television, the cat will sometimes sit on Riley, and sometimes sit on Sam. Riley wonders if Joey has a preference for which person they sit on, or if he chooses to sit on each of them at random. Over the course of a week, during which Riley and Sam binge watched the show *Severance*, they counted how often Joey sat on each of them. At the end of the week, they found that Joey sat on Riley 15 times, and on Sam 5 times.

Let $Y \sim \text{Binomial}(20, \theta)$ denote the number of times Joey sits on Riley in a random sample of 20 occasions when Joey sits on one of them. Consider the discrepancy measure $D = |Y - E[Y; H_0]|$.

(i) [1 mark] Write down the null hypothesis H_0 and give the numeric value (not expression) of $E[Y; H_0]$ (the expected value of Y if the null hypothesis is true).

$H_0 : \theta = 0.5 \qquad E[Y; H_0] = 10$

0.5 marks each.

(ii) [2 marks] Give a numeric expression (not a probabilistic expression) for the p-value resulting from the hypothesis test using the discrepancy measure D and the null hypothesis you gave in (i). Your answer should clearly include a numeric value for d (the observed value of the discrepancy measure).

$d = |15 - 10| = 5$.

$p - \text{value} = P(D \geq d; H_0) = P(D \geq 5; H_0) = \sum_{y=0}^{5} P(Y = y; H_0) + \sum_{y=15}^{20} P(Y = y; H_0)$ where $P(Y = y; H_0) = \binom{20}{y} 0.5^y (1 - 0.5)^{20-y}$

0.5 marks for correct value of d. 0.5 marks for $P(D \geq d)$. 1 mark for the numeric expression.

(iii) [1 mark] Is this an exact or approximate hypothesis test?

Exact.

Reminder: We call an hypothesis test 'exact' when there are no approximations used in its derivation. If we use approximations (such as the central limit theorem) it would be an approximate test.

(iv) [1 mark] You are told that the p-value of the hypothesis test is less than 0.05. Does this prove Joey has a preference for who he sits on? Briefly justify your answer.

No. A hypothesis test does not prove whether a hypothesis is true or false, it shows that the data are inconsistent/unlikely if the null hypothesis is true.

Note: As we have emphasized regularly in class, how we phrase the conclusions of hypothesis tests is very important, and in particular we should avoid definitive statements such as to say the hypothesis is true, or proven.

Part B [15 marks]

Part B begins on the following page and are based on the assignments. All variate names refer to those in the Assignment Dataset Information file. All R commands assume the assignment dataset has been stored in R in an object named `mydata`. You may further assume that the variate `subject.age.log` has been created (as a variate within `mydata`) as described in Analysis 2 of Assignment 1, and that the MASS library has been loaded.

Question 2 [15 marks]

(a) [8 marks] All questions in part (a) concern Assignment 2. Any references to specific analyses, or models, are in relation to those detailed in Assignment 2.

Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) We wish to use R to generate a Q-Q plot of a variate `y` that also includes a straight line for reference. Which of the following commands should we use?

A `qqnorm(y)` only.

B `qqline(y)` only.

C `qqnorm(y)` and `qqline(y)`.

D `qqnorm(y, add = TRUE)` only.

(ii) Let $Y \sim \text{Exponential}(34.56)$. What R command would return $P(Y \geq 25)$?

A `pexp(25, 34.56)`

B `1 - pexp(25, 34.56)`

C `pexp(25, 1/34.56)`

D `1 - pexp(25, 1/34.56)` Note: This command was relied on in Assignment 2, and was also covered in the Introductory R Tutorial.

(iii) Assignment 2 Analysis 1d asked you to plot a cumulative distribution function curve for the model for `subject.age`. Which of these R functions should be used for this?

A `pnorm()`

B `pexp()`

C `dnorm()`

D `dexp()`

(iv) Assignment 2 Analysis 2d asked you to generate a side-by-side boxplot of `vehicle.year`. Which of the following CANNOT be estimated from the boxplot you generated?

A Median `vehicle.year` in each category of your chosen variate in your chosen city.

B Number of observations in each category of your chosen variate in your chosen city.

C Interquartile range of `vehicle.year` in each category of your chosen variate in your chosen city.

D The maximum value of `vehicle.year` in your chosen city.

(v) The following R command is proposed to generate a grouped barplot of a set of observed and expected frequencies (stored as `ob` and `ex`, respectively):

```
barplot(rbind(ob, ex), beside = T, legend("topright", legend = c("Observed", "Expected")))
```

Which of the following should NOT appear within the above `barplot()` command?

A `rbind(ob, ex)`

B `beside = T`

C `legend("topright", legend = c("Observed", "Expected"))`

D All of A-C should appear within the above `barplot()` command.

Questions (vi)-(viii) are each worth one mark. Consider each of the following statements based on Assignment 2. For each statement, discuss whether you agree or disagree, providing a brief justification for your answer in 1-2 sentences.

(vi) "The study population is a subset of the target population."

Disagree. The target population is stops in 2026. The study population is stops prior to this.

Note: This was explored in Assignment 2, Analysis 3.

(vii) "The sample mean and sample standard deviation for `subject.age` are very different, therefore an Exponential model for this variate is not appropriate."

Agree. They are very different when we would expect them to be the same if the data were generated from an Exponential distribution.

(viii) "We could use the comparison of observed and expected frequencies in Assignment 2 Analysis 2a to assess the fit of a Poisson model for the chosen variate."

Disagree. We are comparing observed and expected frequencies in the population, not the Poisson model.

(b) [7 marks] All questions in part (b) concern Assignment 3. Any references to specific analyses, or models, are in relation to those detailed in Assignment 3.

Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) Assignment 3 Analysis 1 fits a Binomial model concerning stops occuring on weekdays and weekends. Based on this analysis, what is the maximum likelihood estimate of the probability a randomly chosen stop takes place on a weekday (not a weekend)?

A 0

B 0.21

C 0.79 [Note: mle of the probability of a weekend stop is 0.21, so for weekday it is 1-0.21]

D 1

(ii) The 10% likelihood interval in Assignment 3 Analysis 1f is [0.182, 0.239]. Suppose you used `uniroot()` to find the upper bound of this interval. What values should we set the options `lower` and `upper` to?

A `lower = 0.16, upper = 0.26`

B `lower = 0.22, upper = 0.26` [Note: This was covered on Assignment 3 and the associated R Tutorial.]

C Both A and B would be appropriate.

D Neither A nor B would be appropriate.

(iii) Which of these commands would be most useful in identifying a 10% likelihood interval when plotting a relative likelihood function?

A `abline(v = 0.1)`

B `abline(v = 10)`

C `abline(h = 0.1)`

D `abline(h = 10)`

(iv) Assignment 3 Analysis 2d asks for a 90% confidence interval for the mean μ in a $G(\mu, \sigma)$ model for `subject.age.log` in San Antonio. Which of the following commands would be useful when constructing this confidence interval?

A `qnorm(0.95)`

B `qt(0.95, 633)` [Note: $n-1$ degrees of freedom.]

C `qt(0.95, 634)`

D `qt(0.95, 949)`

(v) An approximate 90% confidence interval for θ in our Exponential model for `subject.age` is [32.30, 36.81]. If we repeat this analysis using age measured in months instead of years, the resulting interval is [387.6, 441.7]. What is the approximate coverage of this wider interval for mean `subject.age` in months?

A Less than 90%

B 90% [Note: This is also an approximate 90% CI.]

C More than 90%

D Cannot be determined.

Questions (vi)-(vii) are each worth one mark. Consider each of the following statements based on Assignment 3. For each statement, discuss whether you agree or disagree, providing a brief justification for your answer in 1-2 sentences.

(vi) "If ω denotes the probability a randomly chosen stop in the study population happens at the weekend, then based on the assignment dataset $\frac{2}{7}$ is a plausible value for ω."

Disagree. $R\left(\frac{2}{7}\right) = 5.602 \times 10^{-7}$, which is very small (or, equivalently, $\frac{2}{7}$ is outside the 10% likelihood interval for ω).

Note: This was covered on Assignment 3, Analysis 1.

(vii) "We should be concerned about the validity of the Central Limit Theorem approximation when we estimate σ in the $G(\mu, \sigma)$ model in Analysis 2e."

Disagree. We do not use the CLT when estimating σ in this context.

Part C [30 marks]

Question 3 [10 marks]

A manager in a large software company wanted to estimate the mean length (in minutes) of lunch breaks taken by all current employees. To investigate, a random sample of 500 employees were sent a survey via email on 11 November 2025. The survey asked each employee to estimate how long, on average, they spent on their lunch break. The survey was voluntary, and 25 employees responded to the survey. Note that, for the purposes of this question, you may assume that 'current employees' refers to those employed by the company on 11 November 2025.

(a) [3 marks] Would study error be a concern in this study? Carefully justify your answer. If relevant to your answer, the target population, study population, and/or sampling protocol should be described.

The target population is all current employees. The study population is the 500 employees who were randomly chosen to receive the survey. Because it is a random sample we would not expect study error to be a problem, as any attribute of interest should be similar in the target and study populations.

1 mark for target population, 1 mark for study population, 1 mark for noting no study error because the study population is a random sample of the target population.

Note: Please carefully review your grading for this part (and part (b)). These questions are particularly challenging to grade, and if you feel your answer was under-marked then please submit a remark request!

(b) [3 marks] Would sample error be a concern in this study? Carefully justify your answer. If relevant to your answer, the target population, study population, and/or sampling protocol should be described. [Note: If you described any of these in part (a) it is not necessary to describe them again in your answer to part (b).]

The sampling protocol is that only those who voluntarily complete the study enter the sample.

If employees who took longer lunch breaks were less likely to complete the survey than those in the study population, then this could constitute sample error.

1 mark for sampling protocol, 1 mark for discussing difference between study population and sample, 1 mark for linking this to an attribute.

Note: Please carefully review your grading for this part (and part (a)). These questions are particularly challenging to grade, and if you feel your answer was under-marked then please submit a remark request!

(c) [1 mark] Would measurement error be a concern in this study? Briefly explain why or why not.

If employees do not tell the truth about their responses, this would constitute measurement error.

Assume that the length of lunch break for a randomly chosen employee follows a $G(\mu, \sigma)$ distribution. The results of this first survey gave a 95% confidence interval for μ that was 21 minutes wide. However, the manager would like a 95% confidence interval for μ whose width is no greater than 10 minutes.

You have been asked to design a new study where the width of the resulting 95% confidence interval will be no greater than 10 minutes. You may assume that in this new study the standard deviation equals 50. Let n denote the sample size for this new study. (Note: This n equals the number of people who respond to the survey in a future study.)

(d) [1 mark] Write down an equation that gives the width of a 95% confidence interval for this new study as a function of n. You may assume $P(-1.96 \le Z \le 1.96) = 0.95$, where $Z \sim G(0, 1)$.

$$2 \times 1.96 \frac{50}{\sqrt{n}}$$

(e) [2 marks] Calculate the <u>smallest</u> sample size n that will result in a 95% confidence interval of width no greater than 10. (You are reminded you may assume the standard deviation is 50.) We recommend giving a numeric value as your final answer. However, you may leave your answer as a numeric expression if you wish, but it must clearly indicate the *exact* smallest sample size that will yield the confidence interval with the desired width constraint.

$$2 \times 1.96 \frac{50}{\sqrt{n}} \ge 10 \implies 1.96 \frac{50}{\sqrt{n}} \ge 5$$
$$n \ge \left(\frac{1.96 \times 50}{5} \right)^2 = 19.6^2$$

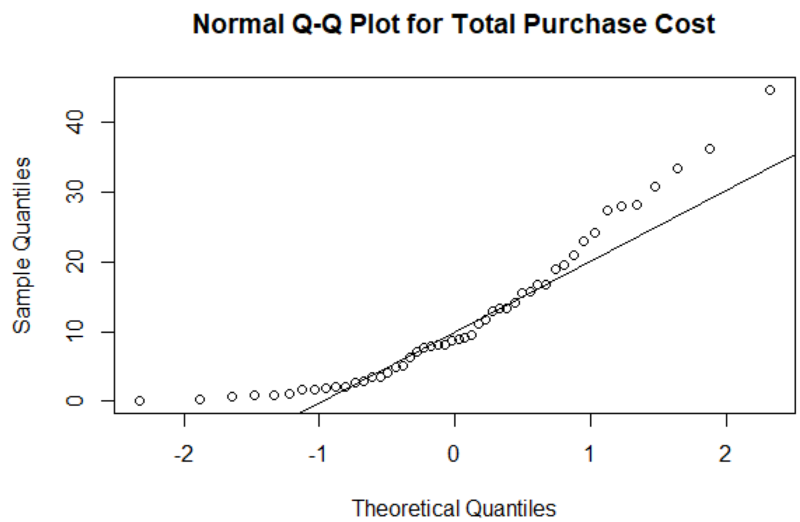
So $n \ge 19.6^2$ or $n \ge 384.16$ so $n = 385$.

Note: For full credit an indication of a specific number is required, as stated in the qustion. While a numeric value was not required, the answer could have been written as (e.g.) "The smallest integer larger than 19.6^2" or $\lceil 19.6^2 \rceil$.

Question 4 [10 marks]

Note: This question spans 3 pages.

Alex and Blake are working on co-op at a Canadian cartoon-themed dollar store called *Loonie Toonies*. They have been asked to investigate the amount of money an individual spends in a single visit to the store. A study is conducted by recording the total spend (to the nearest cent) for each person in a random sample of 50 customers at the store. A Q-Q plot of the money spent is provided below.



(a) [2 marks] Based only on the Q-Q plot, would a Gaussian model be appropriate for these data? Briefly justify your answer, which should highlight any features of the data we can identify from this Q-Q plot that would be important to note in choosing an appropriate distribution.

No, a Gaussian model would not be appropriate. We see a long right tail/evidence of positive skewness, which should be taken into account in choosing a distribution.

Note: Mentioning skewness was required for full marks as it is a feature of the data that would be important to note in choosing an appropriate distribution.

(b) [1 mark] Estimate the sample median of money spent per customer to the nearest dollar.

$\hat{m} = 9$, but any of 8, 9 , or 10 would receive credit.

You assume the following model: Let Y_i represent the total money spent in a single visit to the store by a randomly selected individual and assume $Y_i \sim Exp(\theta)$, $i = 1, \ldots, 50$ independently. You are told $\bar{y} = 11.98$. You wish to test the hypothesis that the true mean for the total spend is \$10 using the test statistic $D = \frac{|\tilde{\theta} - \theta_0|}{\frac{\theta_0}{\sqrt{n}}}$.

You are given the following R commands and output:

```
> pt(-1.400, 49)        > pt(1.400, 49)
[1] 0.08390666          [1] 0.9160933
> pnorm(-1.400)         > pnorm(1.400)
[1] 0.0807567           [1] 0.9192433
```

(c) [1 mark] Briefly explain why the chosen test statistic $D = \frac{|\tilde{\theta} - \theta_0|}{\frac{\theta_0}{\sqrt{n}}}$ is appropriate for this test in 1-2 sentences.

D is appropriate because observed data that gives estimates of θ close to 10 give the smallest possible values of D, and D increases the further $\hat{\theta}$ is from θ_0 in either direction, or increases the less consistent the data is with what we'd expect if the null hypothesis was true.

Reminder: When forming discrepancy measures we often look for two features: (i) Larger values indicate greater inconsistency with the null hypothesis; and (ii) Its distribution can be used to calculate probabilities when the null is true.

(d) [3 marks] Test the null hypothesis $H_0 : \theta = 10$. You are told that for these data $d = 1.4$. You answer should include the following:

- Calculate the p-value for this test, showing the steps you take.
- Make a conclusion for the hypothesis test **within the context of this study**.

$$\begin{aligned}
\text{p-value} &= P(D \geq d; H_0 \text{ true}) \\
&\approx P(|Z| \geq 1.400) \quad Z \sim G(0,1) \\
&= 2P(Z \leq -1.4) \quad (\text{or } 2[1 - P(Z \leq 1.4)]) \\
&= 0.162
\end{aligned}$$

There is no evidence against the null hypothesis that the mean purchase cost is \$10 at Loonie Toonies.

Reminder: When we use a test statistic based on a Central Limit Theorem approximation, we refer to the $G(0, 1)$ distribution to calculate the p-value.

You are told that the relative likelihood function for the Exponential model is

$$R(\theta) = \left(\frac{\bar{y}}{\theta}\right)^n e^{-n\left(\frac{\bar{y}}{\theta}-1\right)}$$

and given the following R commands and output:

```
> pchisq(1.735, 1)        > pchisq(1.735, 2)
[1] 0.8122264/            [1] 0.5799998
> pnorm(-1.735)           > pnorm(1.735)
[1] 0.0413704             [1] 0.9586296
```

(e) [2 marks] Show that the observed likelihood ratio statistic for this study can be calculated by $\lambda(\theta_0) = -100 \log\left(\frac{\bar{y}}{\theta_0}\right) + 100\left(\frac{\bar{y}}{\theta_0} - 1\right)$.

$$\begin{aligned}
\lambda(\theta_0) &= -2\log R(\theta_0) \\
&= -2\log\left(\frac{\bar{y}}{\theta_0}\right)^{50} - 2(-50)\left(\frac{\bar{y}}{\theta_0} - 1\right) \\
&= -100\log\left(\frac{\bar{y}}{\theta_0}\right) + 100\left(\frac{\bar{y}}{\theta_0} - 1\right)
\end{aligned}$$

1 mark for $\lambda(\theta_0) = -2\log R(\theta_0)$. 1 mark for showing simplification.

(f) [1 mark] You are told that $\lambda(10) = 1.735$. What p-value results from the test of the null hypothesis?

$$\begin{aligned}
\text{p-value} = P(\Lambda(\theta_0) \geq 1.735; H_0) &= P(W \geq 1.735) \quad W \sim \chi^2(1) \\
&= 1 - 0.8122264 \\
&= 0.1877736
\end{aligned}$$

Question 5 [10 marks]

A statistics professor has discovered a new probability distribution which they have named the Gromit distribution. The Gromit distribution has range $y > 0$ and is parameterized by an unknown parameter $\theta > 0$.

Let $Y_1, Y_2, \ldots, Y_n$ denote independent and identically distributed random variables that each follow the Gromit distribution. You are told that

$$V = \frac{4\overline{Y}}{\theta} \sim \chi^2_n$$

is an exact pivotal quantity for θ, where $\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$.

A sample $y_1, y_2, \ldots, y_{30}$ is randomly drawn from the Gromit distribution. You are told that $\overline{y} = 2$ with sample standard deviation 4. You are given the following R commands and output:

```
> qchisq(0.05, 29)    > qchisq(0.1, 29)    > qchisq(0.9, 29)    > qchisq(0.95, 29)
[1] 17.70837          [1] 19.76774         [1] 39.08747         [1] 42.55697
> qchisq(0.05, 30)    > qchisq(0.1, 30)    > qchisq(0.9, 30)    > qchisq(0.95, 30)
[1] 18.49266          [1] 20.59923         [1] 40.25602         [1] 43.77297
> qnorm(0.05)         > qnorm(0.1)         > qnorm(0.9)         > qnorm(0.95)
[1] -1.644854         [1] -1.281552        [1] 1.281552         [1] 1.644854
```

(a) [1 mark] Write down values a and b that satisfy $P(a \le V \le b) = 0.9$ when $n = 30$.

$a = 18.49266$ $b = 43.77297$ [1 mark]

Alternate answer: $a = 0$ $b = 40.25602$ [1 mark]

(b) [3 marks] Using V, construct a 90% confidence interval for θ based on the observed data. **Carefully justify your answer**, showing all your steps in the derivation of the confidence interval. The bounds of your intervals may be left as numeric expressions. If you were unable to complete part (a), then your answer may involve probabilistic expressions.

1. Find a, b such that

$$P\left(a \le \frac{4\overline{Y}}{\theta} \le b\right) = 0.9$$

Giving the values in part (a).

2. Rearrange to isolate θ: $P\left(a \le \frac{4\overline{Y}}{\theta} \le b\right) = P\left(\frac{a}{4\overline{Y}} \le \frac{1}{\theta} \le \frac{b}{4\overline{Y}}\right) = P\left(\frac{4\overline{Y}}{b} \le \theta \le \frac{4\overline{Y}}{a}\right)$
3. The 90% CI is then $\left[\frac{4\overline{y}}{b}, \frac{4\overline{y}}{a}\right]$ where $n = 30, \overline{y} = 2$, and a and b are as found in part (a).

Note: We have covered many examples like this in class/the Course Notes. If you had difficulty with this question, you may find it helpful to review Tutorial 3 in particular.

(c) [3 marks] The 90% confidence interval found in part (b) would be approximately what level of likelihood interval? Carefully justify your answer: a correct numeric value with no working/justification will not receive full credit. You may use relevant theorems or other results from the course materials, and may assume all required asymptotic results are satisfied.

Find c such that $P(W \leq c) = 0.9$ where $W \sim \chi^2_1$

Equivalently, such that $P(Z^2 \leq c) = 0.9$ where $Z \sim G(0,1)$

This gives $P(-\sqrt{c} \leq Z \leq \sqrt{c}) = 0.9$ giving $c = 1.644854$ from the R output.

The likelihood level is therefore $e^{-c/2} = e^{-1.644854^2/2} = 0.2585227$

Note: As well as examples covered in class, another example of this type of calculation can be found on Assignment 3 for additional practice.

(d) [3 marks] Consider the following random variables, all of which you may assume are independent of one another:

$$V \sim \chi^2_n \qquad W \sim \chi^2_{2n} \qquad Z_1 \sim G(0,1) \qquad Z_2 \sim G(0,1)$$

Consider the following random variables that are constructed using V, W, and Z. For each, give its distribution:

(i) $V + W \sim \chi^2_{3n}$

(ii) $Z_1^2 + Z_2^2 \sim \chi^2_2$ or Exponential(2)

(iii) $\frac{Z}{\sqrt{W/2n}} \sim t_{2n}$

[1 mark each]